

ContextGraph

Literature Survey

To situate ContextGraph, existing approaches to persistent AI context and memory were reviewed across three groups: memory features built into commercial chat products, open agent-memory frameworks and research, and protocol-level or knowledge-graph approaches to portable context. A summary comparison is given first, followed by a short discussion of each group and the resulting research gap.

1. Comparative Summary

#	System / Work	Year	Key Idea	Limitation (relative to this project)
1	ChatGPT Memory (OpenAI)	2024	Model silently extracts and stores user facts across conversations.	Single flat store, one vendor only, no explicit hierarchy or user-triggered save.
2	Claude Projects / Memory (Anthropic)	2025	Project-scoped knowledge plus persistent memory inside Claude apps.	Scoped to Claude only; not reachable by other AI tools; no cross-project graph.
3	Model Context Protocol (Anthropic)	2024	Open JSON-RPC standard letting any AI client call external tools/resources.	Defines the transport/tool-call standard only, not a memory data model.
4	MemGPT (Packer et al., arXiv:2310.08560)	2023	Treats the LLM context window like OS virtual memory, paging facts between core and archival tiers.	A single-agent runtime construct, not a hosted multi-user, multi-client service.
5	Mem0 (arXiv:2504.19413)	2025	Extracts, scores, and retrieves memories for agents with a production-oriented pipeline.	A memory SDK/library; no scoped-hierarchy model or MCP-native distribution built in.
6	Zep (Rasmussen et al., arXiv:2501.13956)	2025	Temporal knowledge graph that tracks how facts about a user change over time.	Strong on temporal graph reasoning; not built around per-user API keys or a visual dashboard.
7	LangChain / LangGraph memory modules	2024	Composable memory abstractions (buffer, summary, vector-store backed) for custom agents.	A toolkit for assembling a bespoke memory system, not an out-of-the-box product.
8	Graph-based PKM tools (Obsidian, Roam)	—	Let a person manually build and browse a linked-note knowledge graph.	Manual curation only; no AI read/write access and no relevance decay.

2. Commercial Memory Features

ChatGPT Memory and Claude's Projects/memory both improved on stateless chat by letting the model retain facts across sessions. Both remain single-vendor: context built up inside one product cannot be read by, or moved to, a different AI tool, which is precisely the situation ContextGraph users are in — they use more than one assistant.

3. Open Agent-Memory Frameworks

MemGPT, Mem0, and Zep each treat memory as an engineering problem inside a single agent runtime: how to page, score, or temporally index facts so an LLM with a bounded context window can still reason over a long history. They are memory *engines* that a developer wires into their own agent — useful building blocks, but none ship as a hosted, per-user service with an API key and a dashboard a non-developer end user can look at.

4. Protocol and Knowledge-Graph Approaches

The Model Context Protocol (MCP) solves a different half of the problem: it standardizes how an AI client calls an external tool or reads an external resource, regardless of which AI vendor built the client. It says nothing about what shape that external memory should take. Graph-based personal knowledge management tools (Obsidian, Roam) show that a linked-node graph is a workable mental model for personal context, but they are built for manual human browsing, not for being queried and updated by an AI agent mid-conversation.

5. Research Gap

No reviewed system combines all of the following: (i) a hierarchical, scope-aware context model that prevents cross-project leakage, (ii) access through a standard, multi-client protocol rather than a single vendor's app, (iii) an explicit, user-visible save and relevance mechanism instead of silent inference, and (iv) a consumer-facing visual graph the user can inspect and edit directly. ContextGraph's contribution is combining these four properties — hierarchy, protocol-level portability, explicit judgment, and visualization — into a single hosted service.

References

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- [7] LangChain. “Memory modules — LangChain / LangGraph documentation.” 2024.
- [8] Obsidian.md and Roam Research — graph-based personal knowledge management products, ongoing.